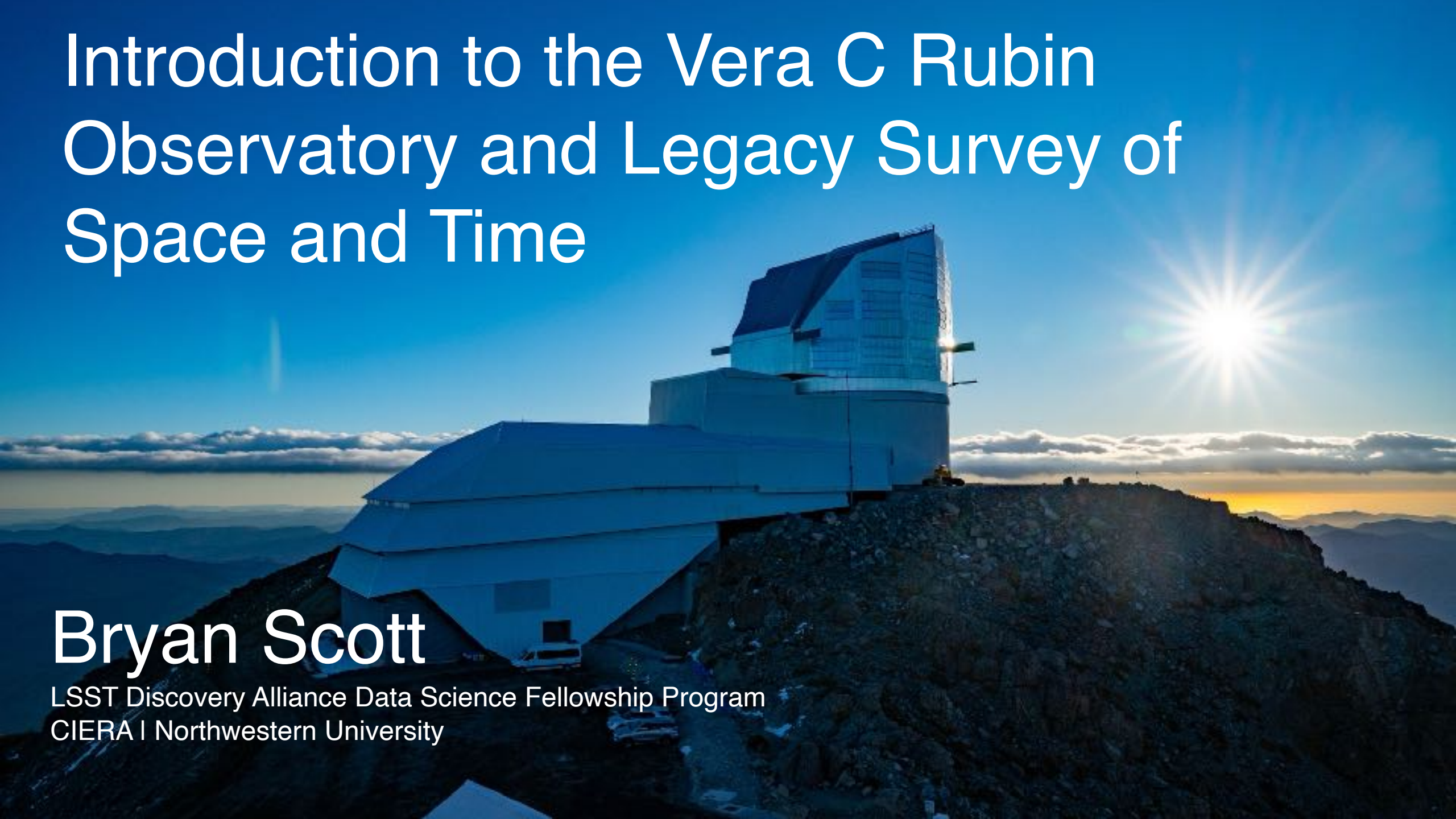


Introduction to the Vera C Rubin Observatory and Legacy Survey of Space and Time



Bryan Scott

LSST Discovery Alliance Data Science Fellowship Program
CIERA | Northwestern University

The DSFP:



“Equipping Astronomers for a Data Rich Future”

Outline of today's talk

Introduction to the Vera C Rubin Observatory: Science Objectives, Construction, Operation, and Data Overview

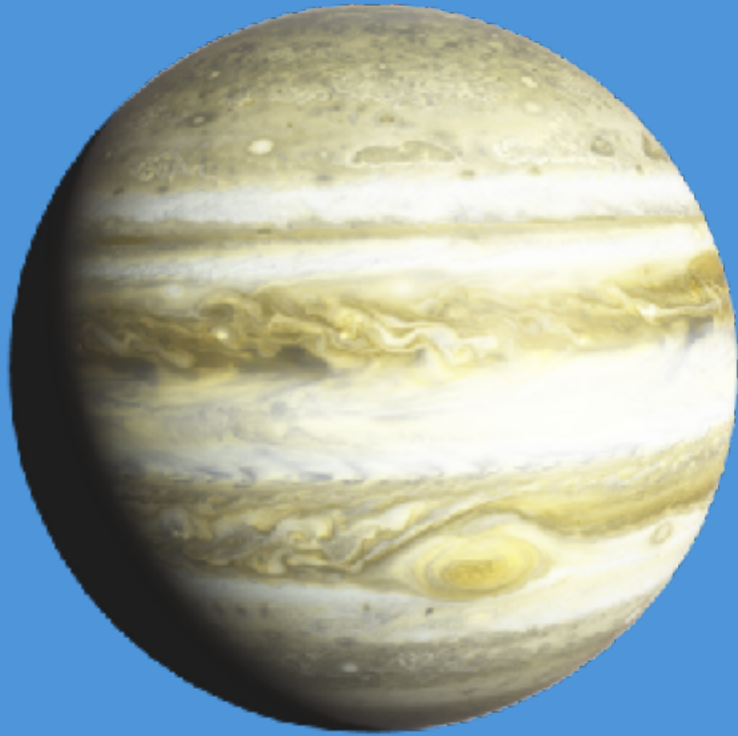
Data Challenges

1. Handling and Communicating (about) Large Datasets
2. Understanding Structure in High Dimensional Datasets
3. Finding Anomalies and Enabling Discovery at Scale
4. Tasking Follow-up Resources

Image attribution note: Throughout this talk many construction and operation images, videos, infographics, and icons are produced by the Vera C Rubin Observatory Project and available in the image and project galleries at rubinobservatory.org

Science Objectives

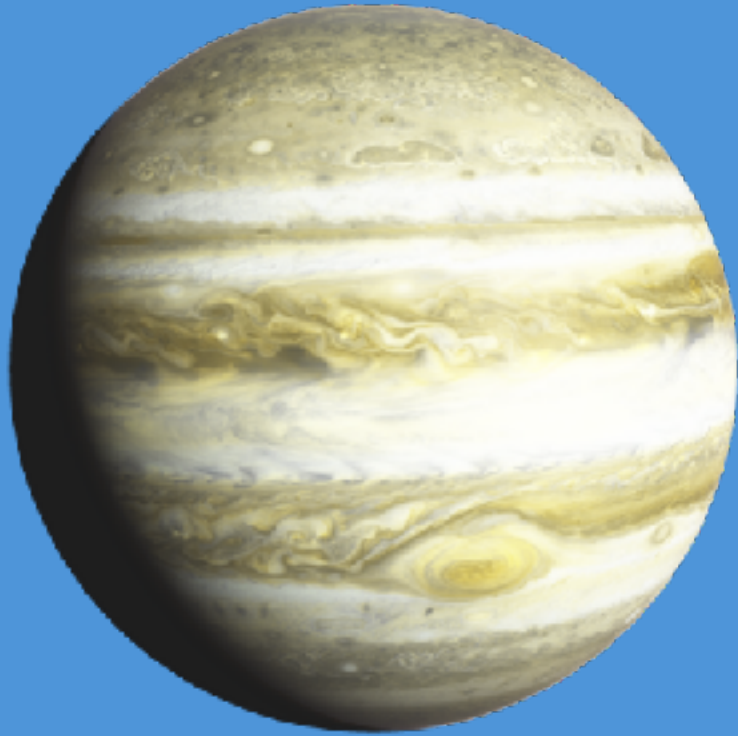
Solar System History and Structure



Characterizing the orbits of up to 90% of Near Earth Asteroids

Measuring orbits of main belt asteroids and trans-Neptunian objects to understand the origin of the planets

Solar System History and Structure



Characterizing the orbits of up to 90% of Near Earth Asteroids

Measuring orbits of main belt asteroids and trans-Neptunian objects to understand the origin of the planets

More speculatively: searching for large outer solar system bodies ("planet 9")

The Variable Universe



Exploding Stars

Transiting Exoplanets and
Microlensing

Variable Star populations

Galaxies (including the Milky Way and AGN)



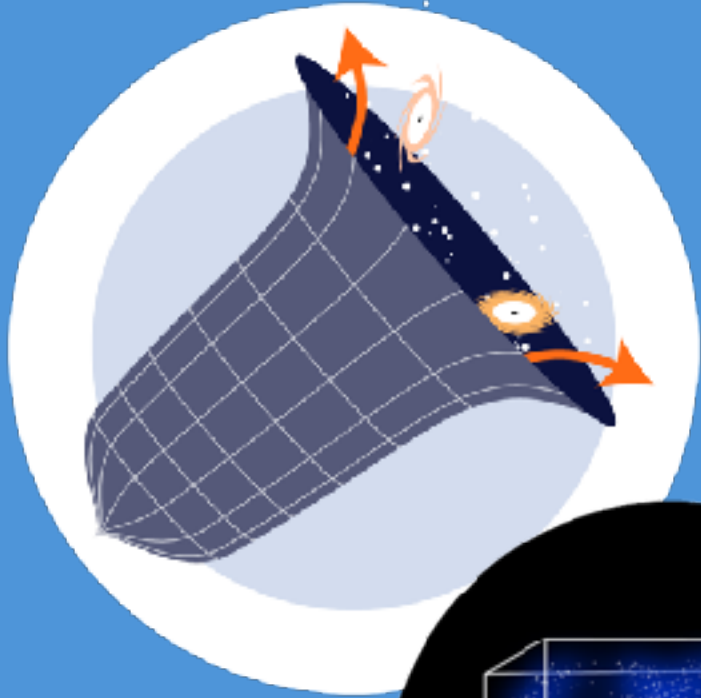
Demographics of the Galaxy
Population | Scaling Relations

Galaxies and their Environments

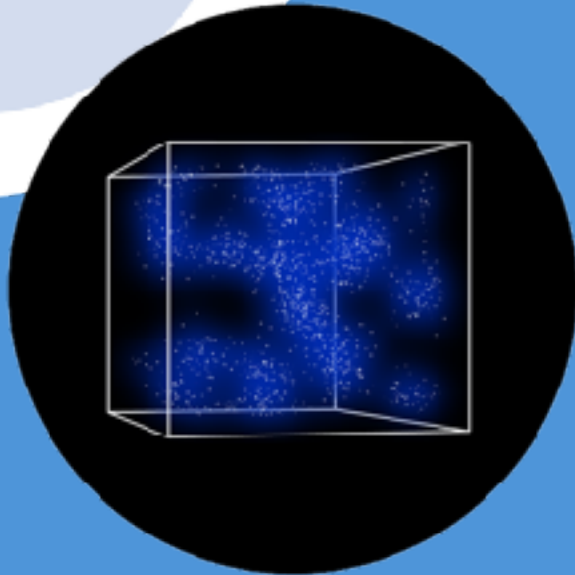
Multi-band mapping of the Milky
Way

AGN luminosity and redshift
relationships

Cosmology and Fundamental Physics

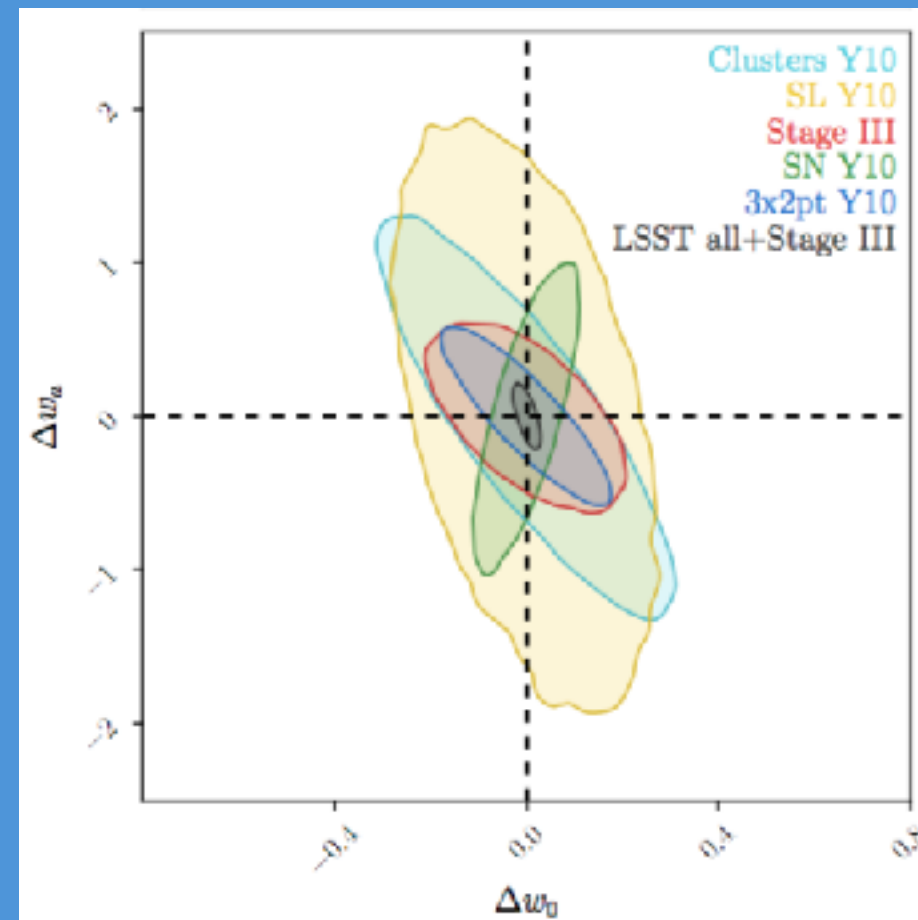
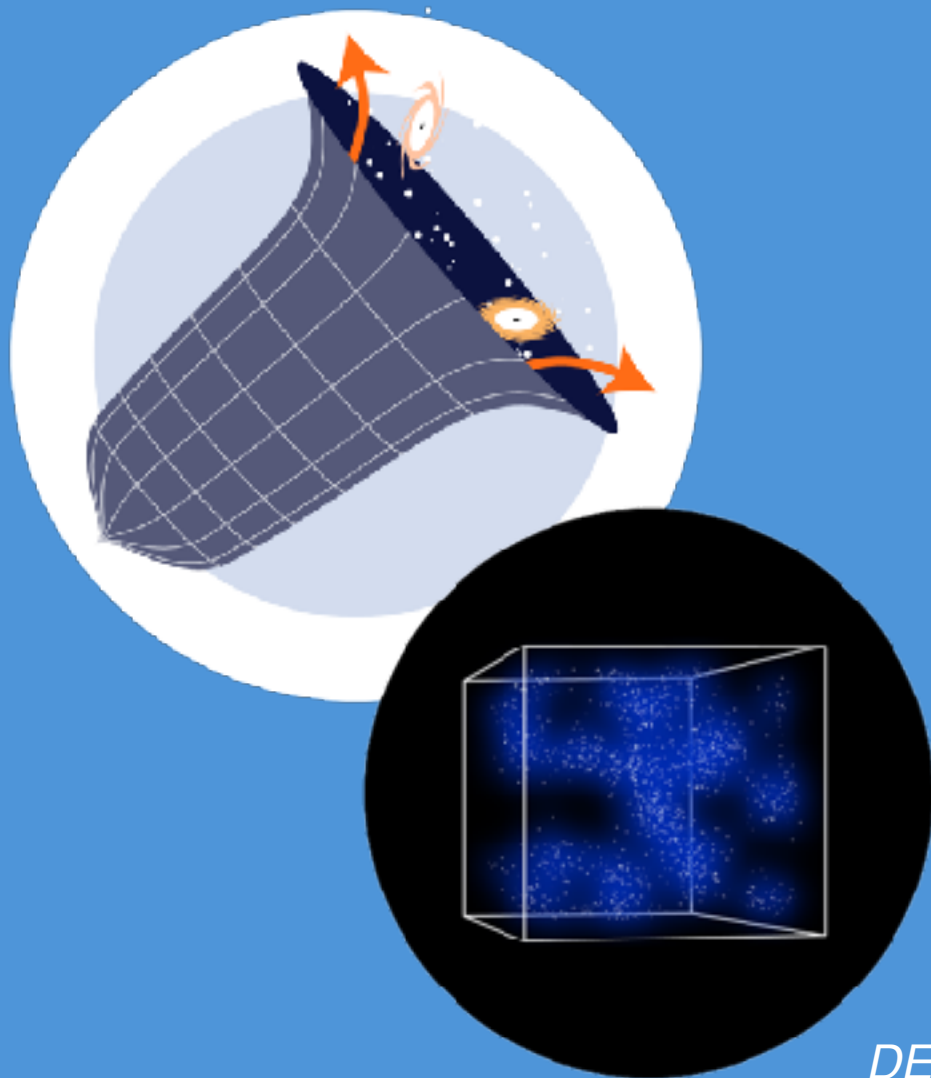


Move beyond a phenomenological model for Dark Matter and Dark Energy



Measure fundamental physics from large scale structure, supernova, and in combination with the CMB.

Example: Dynamical Dark Energy Models



The Vera C Rubin Observatory Legacy Survey of Space & Time

Vera C Rubin Observatory Community



LSST
Discovery
Alliance



Vera C Rubin Observatory Community



LSST
Discovery
Alliance



RUBIN OBSERVATORY

IN-KIND CONTRIBUTIONS



La Serena
Coquimbo

Cerro Pachón

Viña del Mar
Valparaíso

Santiago



By M. Urzúa Zuñiga/International Gemini Observatory - Cerro Pachón from Cerro Tololo, 2006, CC BY-SA 3.0, commons.wikimedia.org/w/index.php?curid=137746188

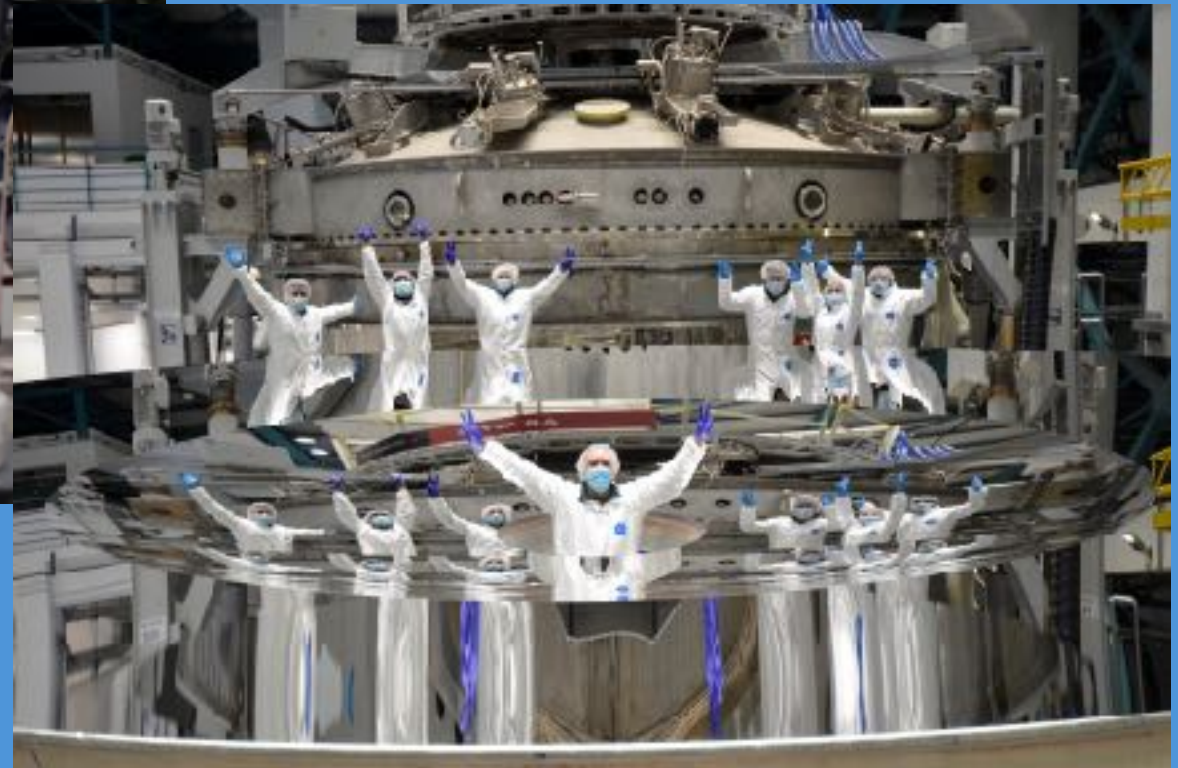
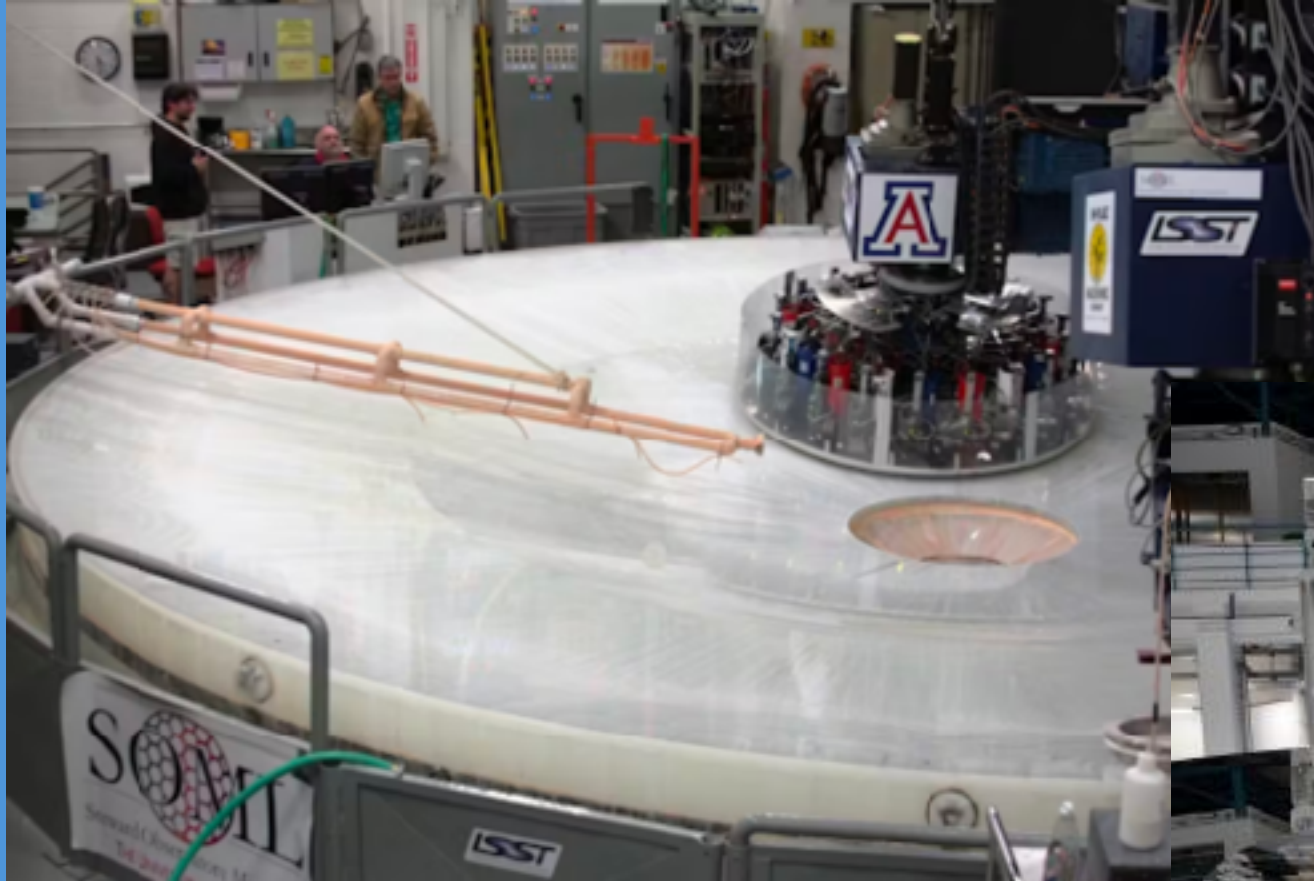
By Puzhok - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=8046529>



An 8.4 meter survey telescope

10 year depths: u : 25.6 g : 26.9 r : 26.9 i : 26.4 z : 25.6 y : 24.8

Rubin Observatory Mirror





An 8.4 meter survey telescope

10 year depths: u : 25.6 g : 26.9 r : 26.9 i : 26.4 z : 25.6 y : 24.8

*Rubin Observatory represents an Order of Magnitude improvement in Entendue
(Product of Area and FOV):*

319 m²deg² (Roman is m²deg²; Palomar/ZTF: 50 m²deg²)

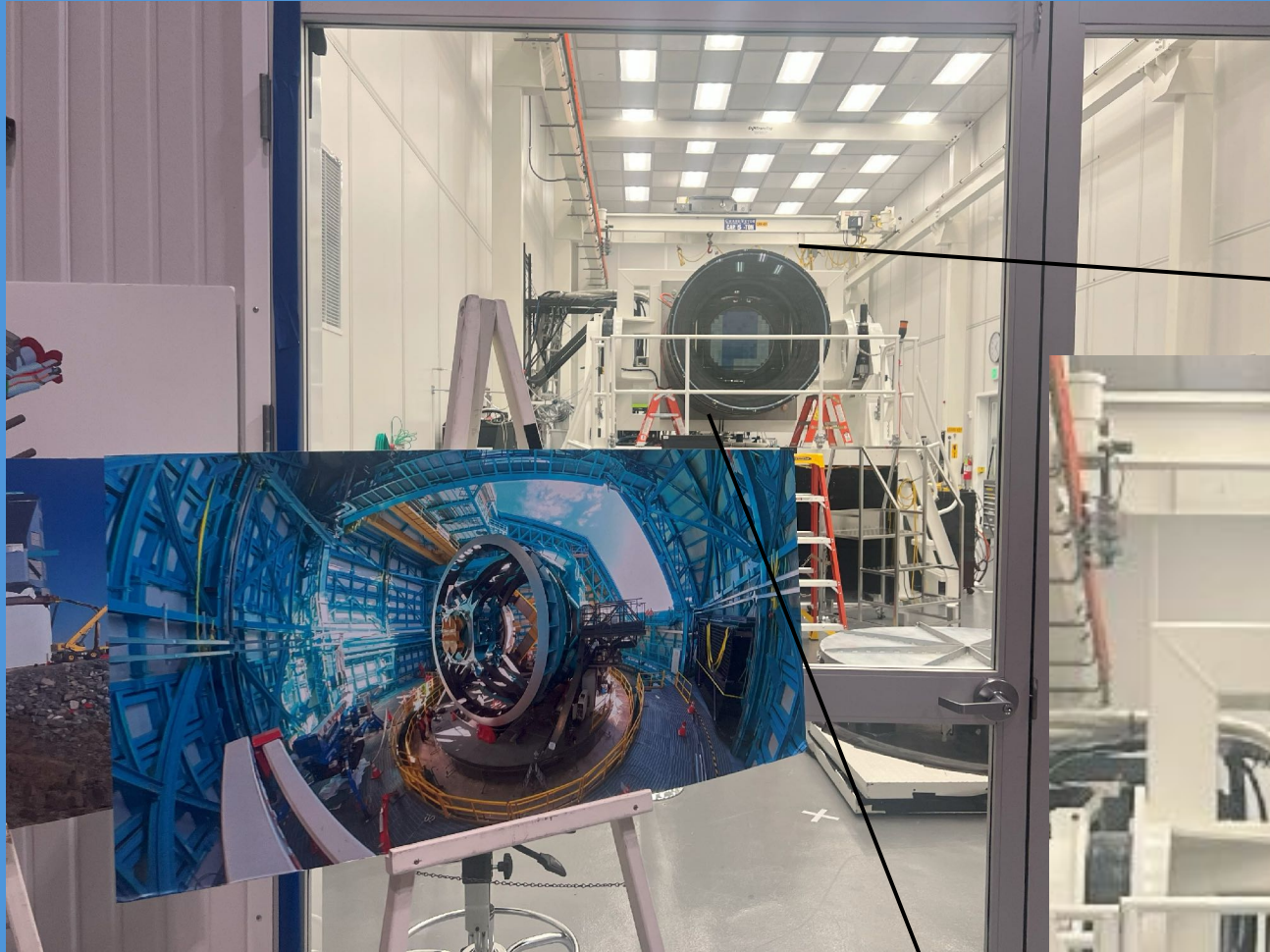
LSST Camera

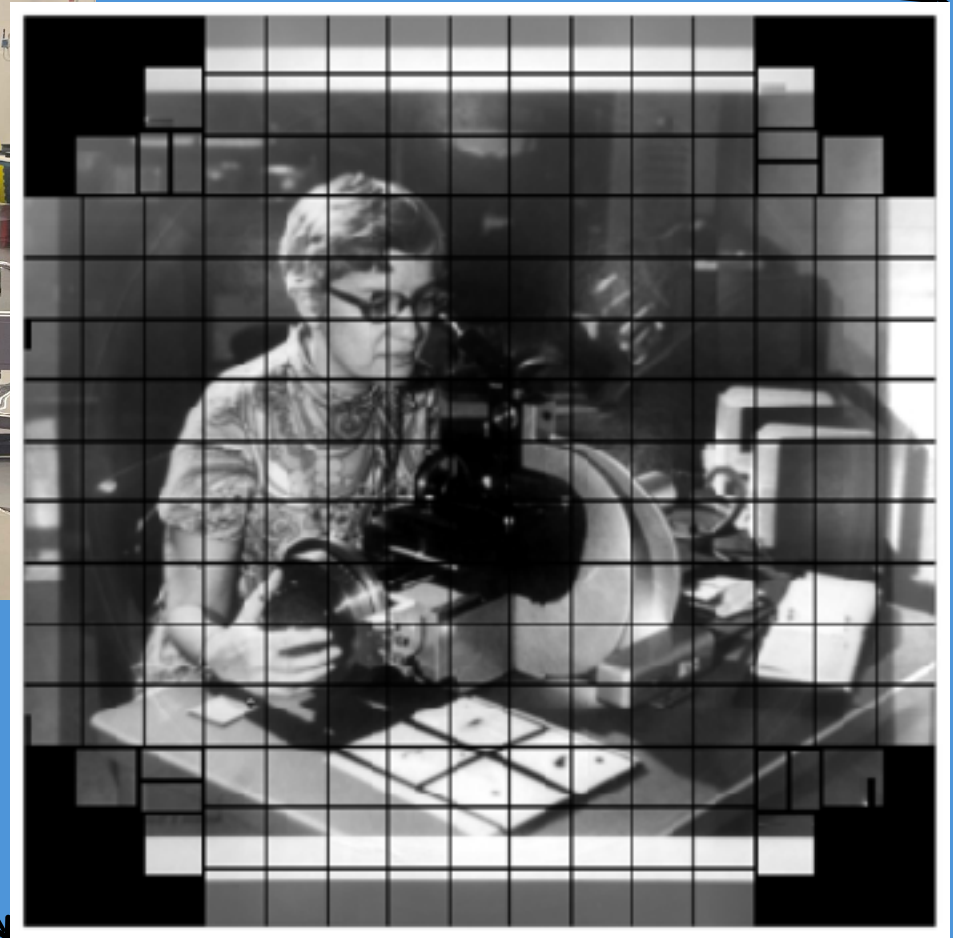
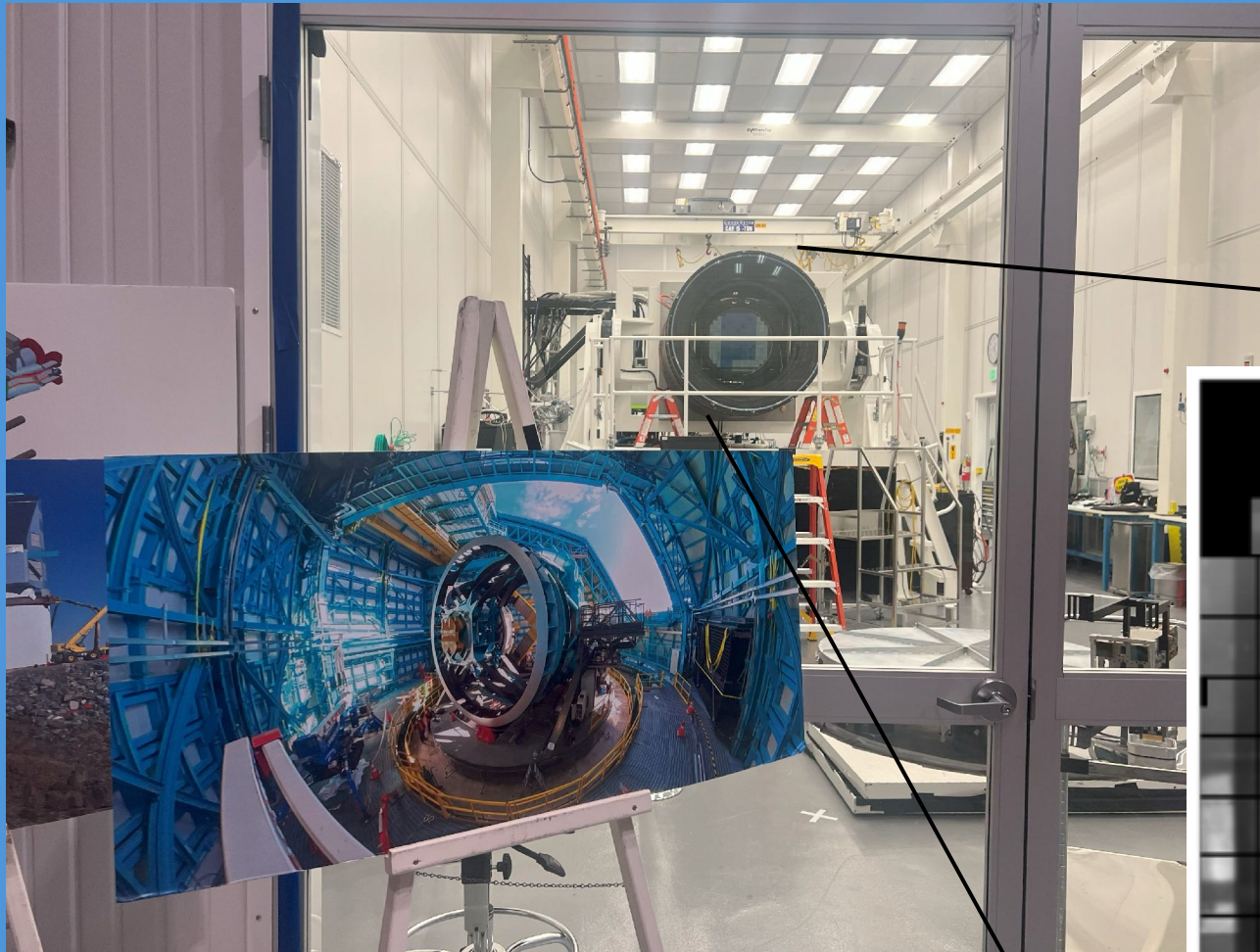


World's Largest Camera:
32 Gigapixels

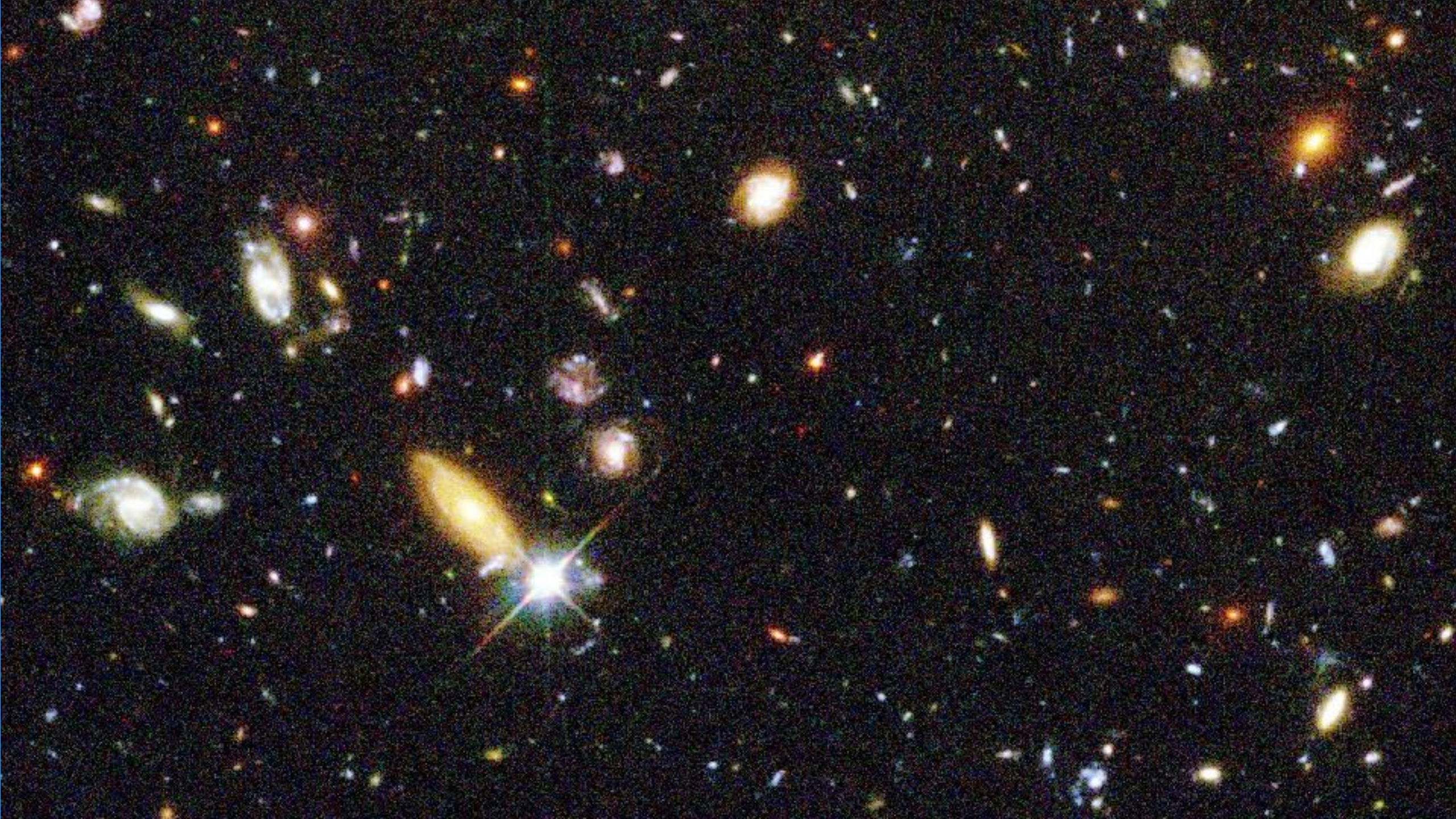
189 16 megapixel sensors
in 21 rafts.

Extremely flat focal plane

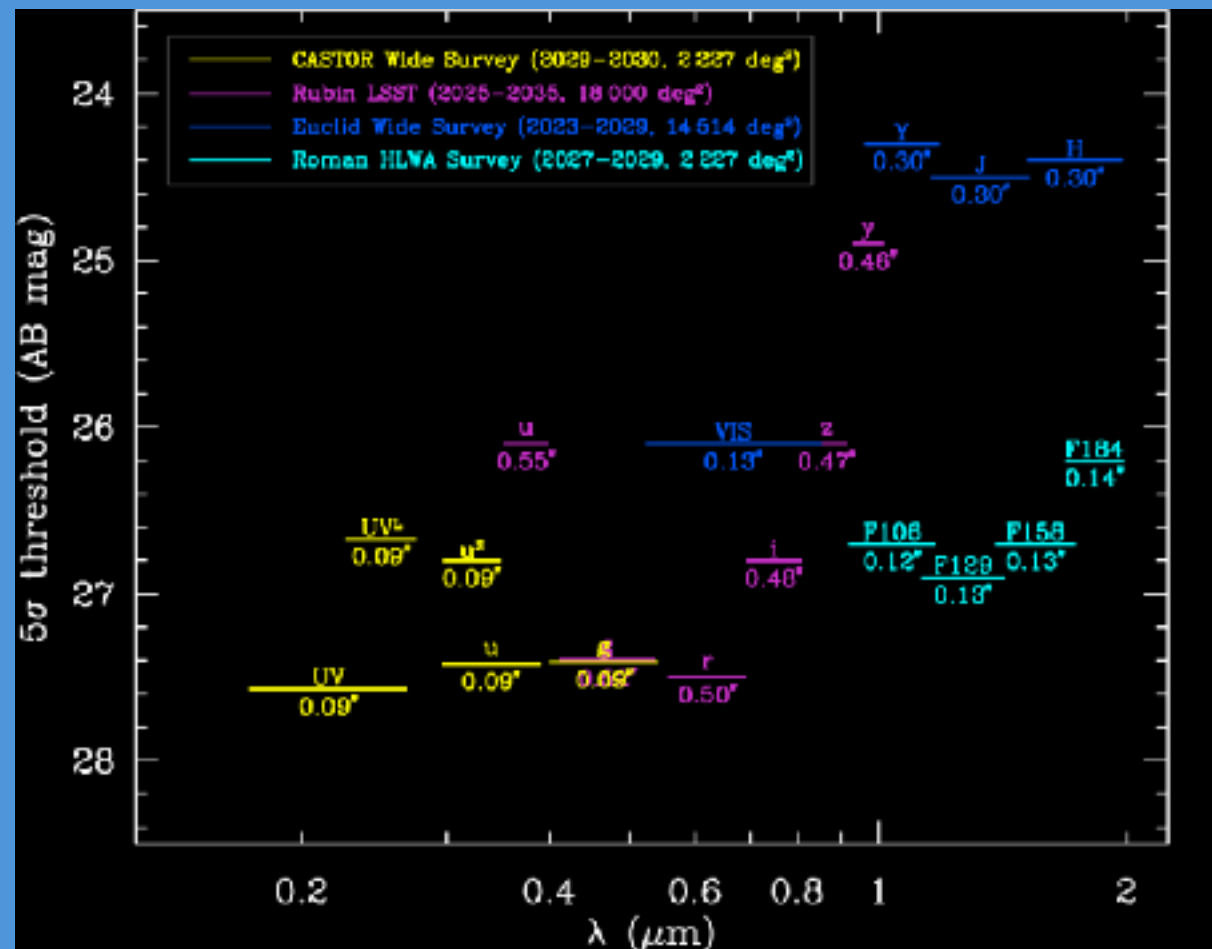




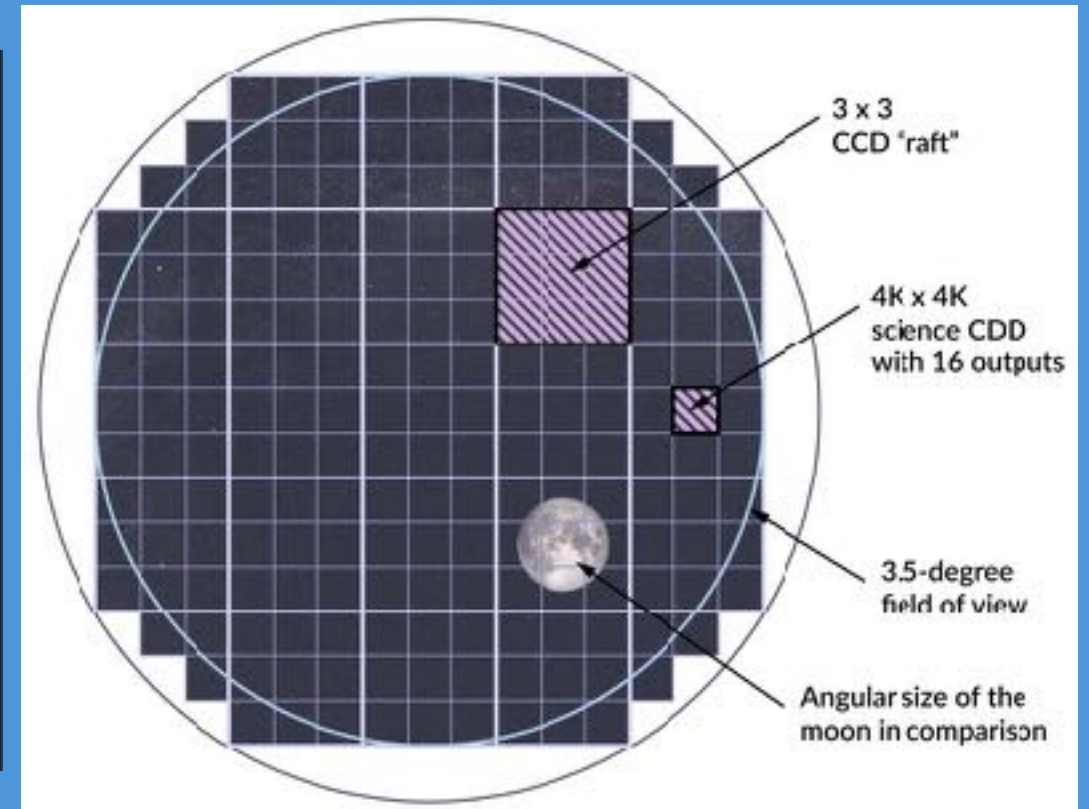




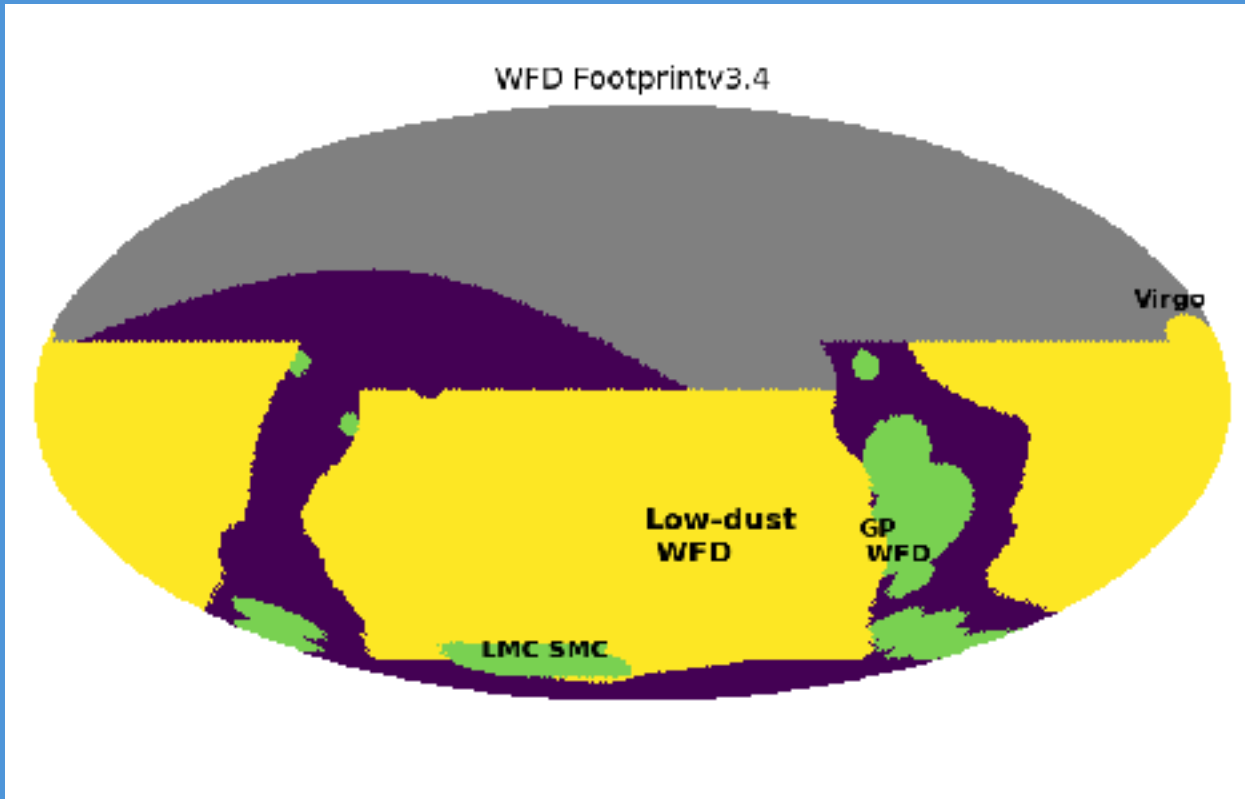
Stage IV Surveys



Stage IV Surveys



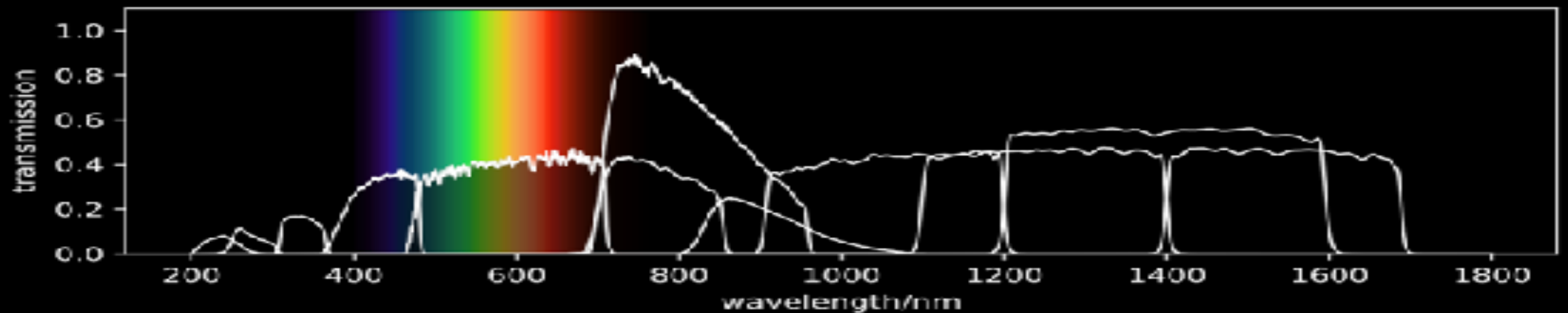
Survey Overview



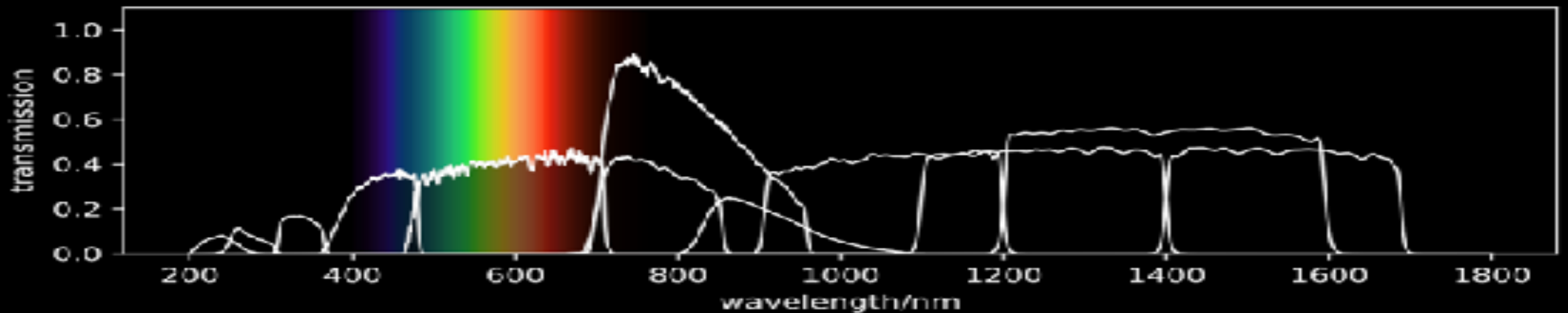
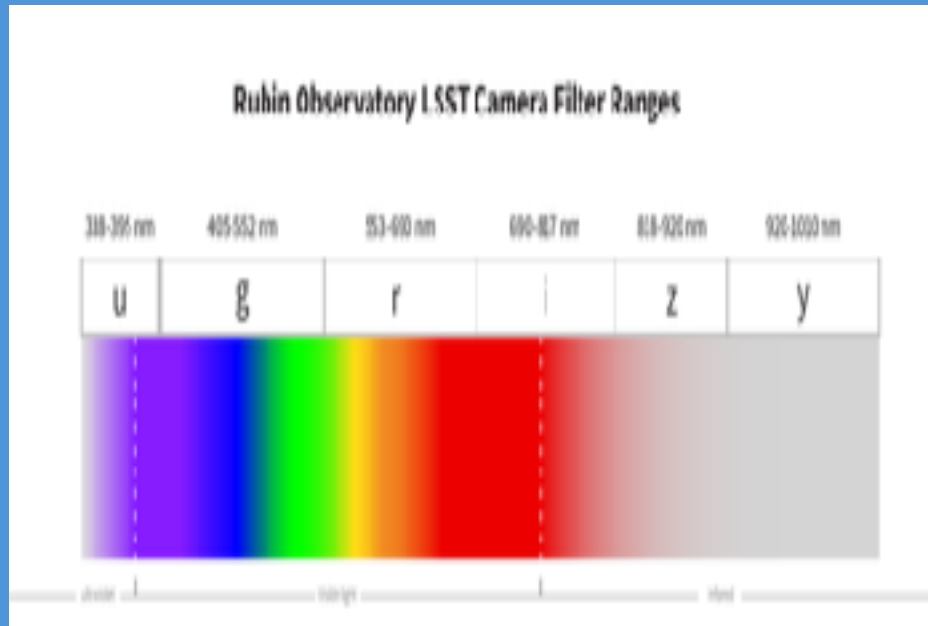
“Wide-fast-deep” survey of
~18,000 sq deg of *southern*
sky

“Deep Drilling Fields”, 10 sq
deg each plus further “mini-
surveys” to support special
science cases

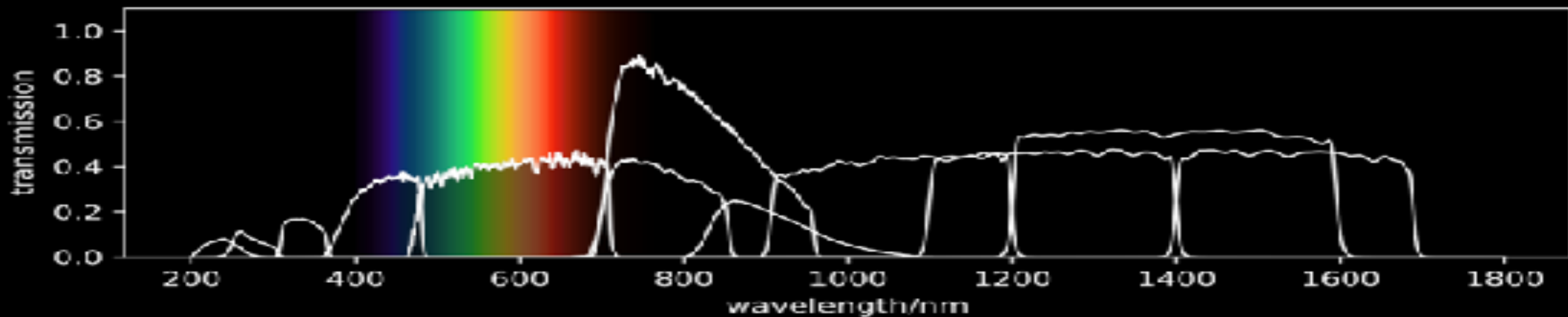
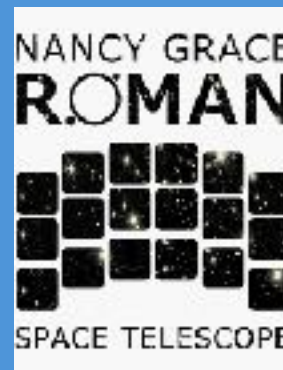
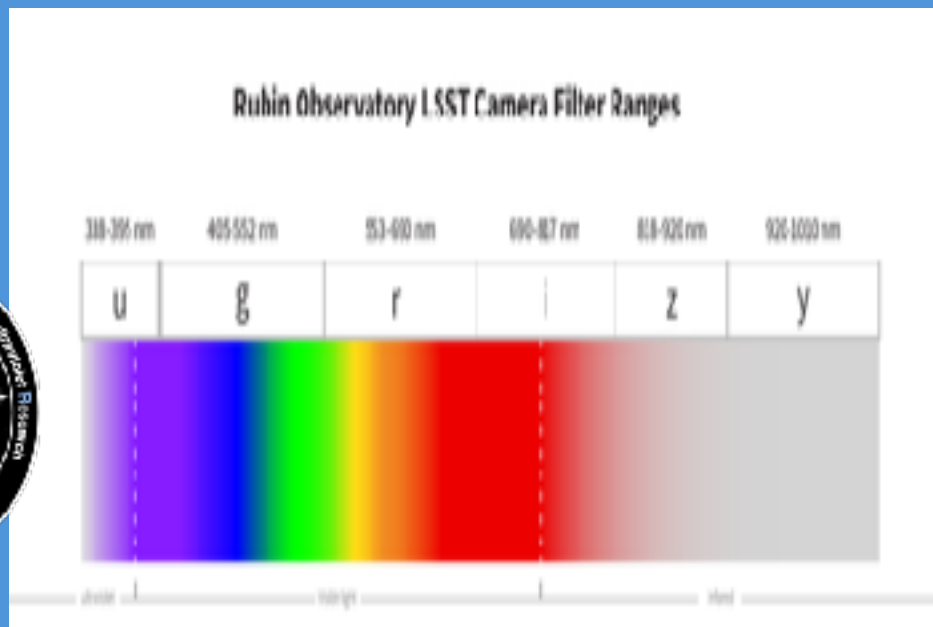
LSST Camera



LSST Camera

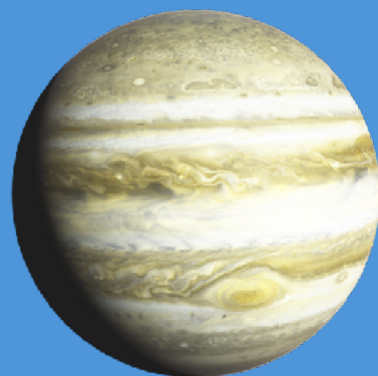


Stage IV Survey Filters

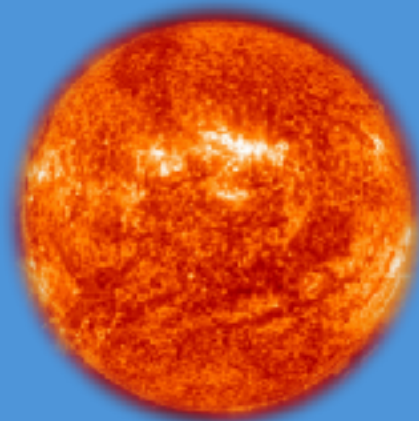




20 billion
galaxies



6 million
orbits



17 Billion
Stars



90% of the
universe's
history

10,000,000 Transients per night

Data Challenges

Challenge 1: Handling and Communicating about Large Datasets



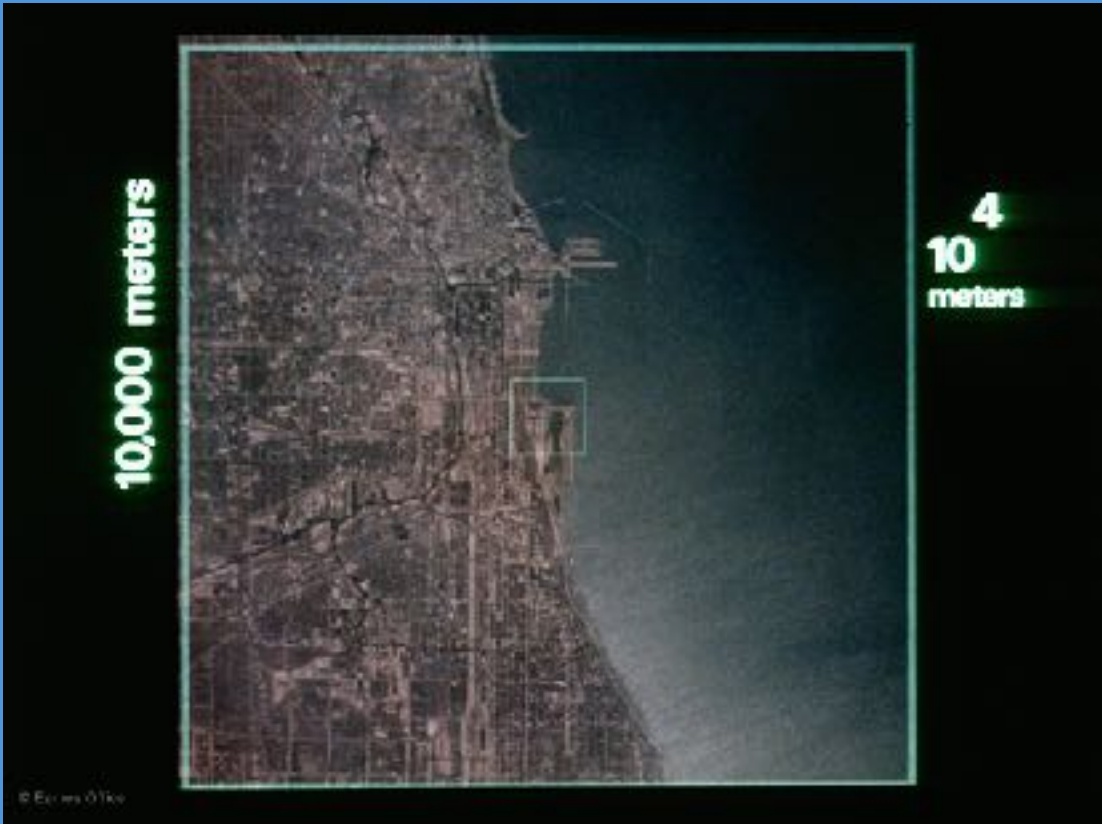
100,000 light years, the Milky Way galaxy, modeled from red-shifted radio waves.

POWERS OF TEN

A FILM DEALING WITH
THE RELATIVE SIZE OF THINGS IN THE UNIVERSE
AND THE EFFECT OF ADDING ANOTHER ZERO

MADE BY THE OFFICE OF
CHARLES AND RAY EAMES

FOR
IBM



Kiloscale (up to $\sim 10^4$ entities)

Megascale ($\sim 10^{5-7}$)

Gigascale ($\sim 10^{8-10}$)



Kiloscale (up to $\sim 10^4$ entities)



Megascale ($\sim 10^{5-7}$)

Gigascale ($\sim 10^{8-10}$)

*LSST Informatics and Statistics
Collaboration*

Kiloscale

Analysis of multiband, multi-epoch photometry for a single object.

*LSST Informatics and Statistics
Collaboration*

Kiloscale

Analysis of multiband, multi-epoch photometry for a single object.

→ Statistical Challenges

*LSST Informatics and Statistics
Collaboration*

Megascale

Demographic analysis of larger populations

population-level re-analysis of previous survey catalogs

multicolor, multi-epoch image analysis

Megascale

Demographic analysis of larger populations

population-level re-analysis of previous survey catalogs

multicolor, multi-epoch image analysis

→ Statistical + Computational Challenges

Gigascale

LSST image data analysis of a single field.

Demographic analysis of very large samples of the 10 year galaxy or star catalogs.

Discovery in the full data releases.

Gigascale

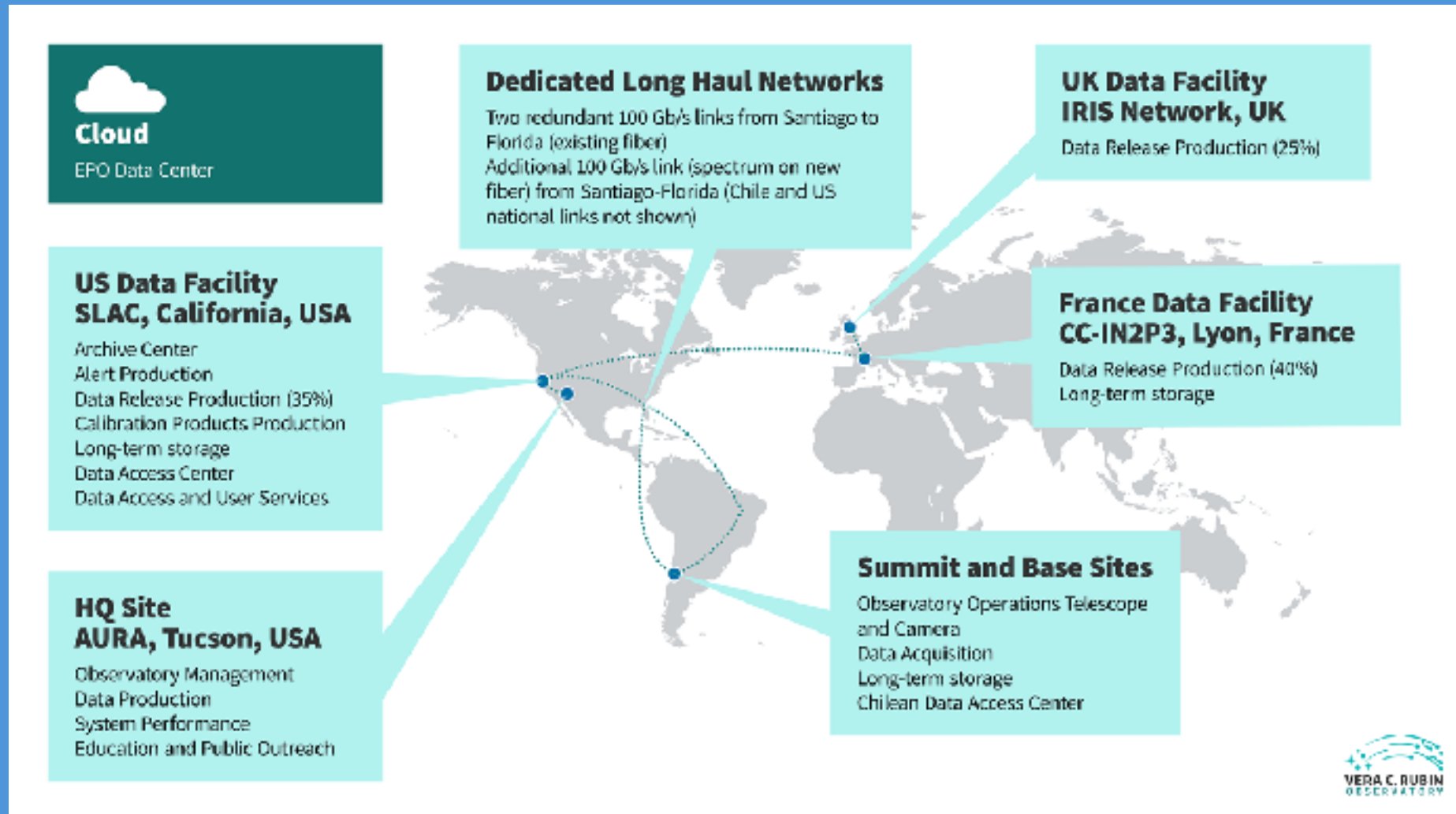
LSST image data analysis of a single field.

Demographic analysis of very large samples of the 10 year galaxy or star catalogs.

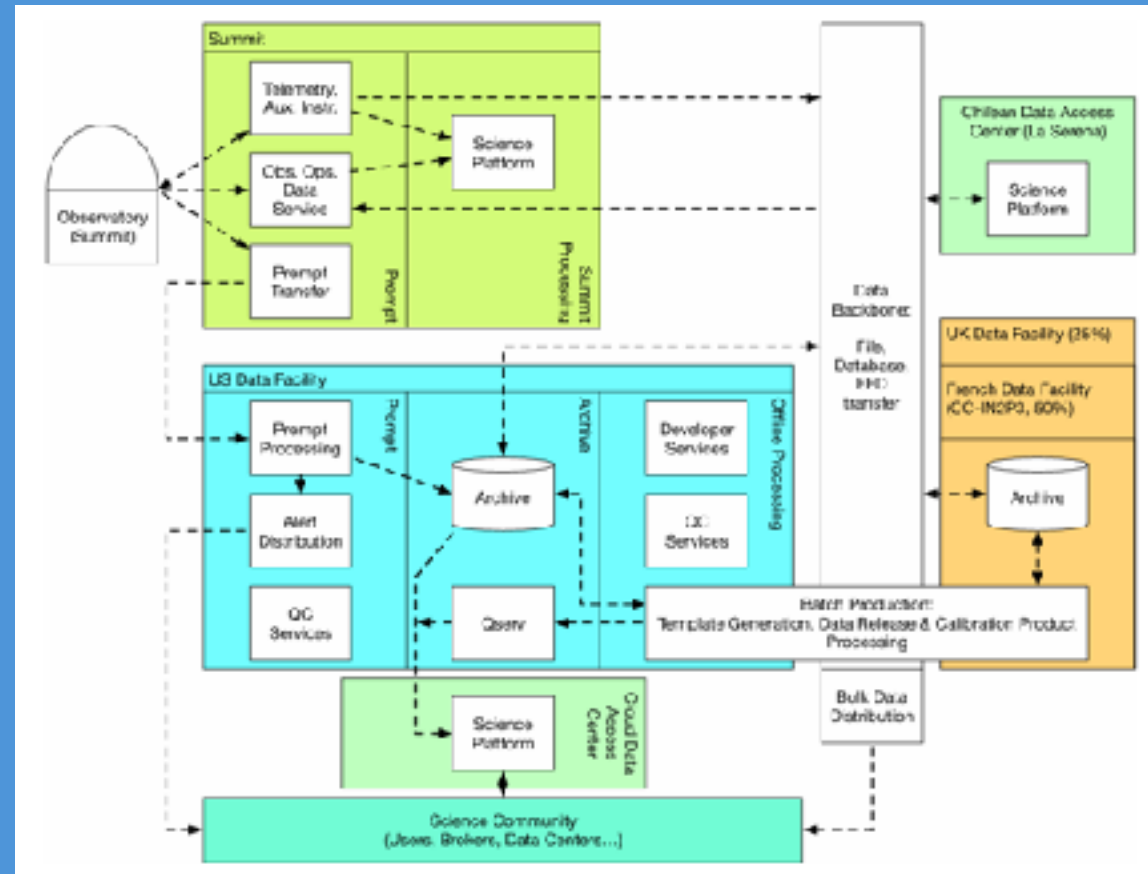
Discovery in the full data releases.

→ Computational Challenges

Data Infrastructure is a Global Effort



Paradigm: Bring Compute and Data Together



Collaboration Level Analyses



LSSTDESC Data Portal

LSST Dark Energy Science Collaboration Public Data Access



Welcome to the **LSSTDESC** Data Portal

If you are new to Globus or have questions, please see the "Getting Started" section below. Otherwise, to download datasets, click [Login](#) above to authenticate into Globus using your organizational login or existing GlobusID. If needed, create a [GlobusID](#) using the "Sign Up" link.

Rubin Science Platform

Portal

Discover data in the browser



[Learn more about the portal.](#)

Notebooks

Process and analyze LSST data with Jupyter notebooks in the cloud



[Learn more about notebooks.](#)

APIs

Learn how to programatically access data with Virtual Observatory interfaces



Democratizing Science

“Open data alone is not sufficient for inclusivity.”

Democratizing Science

“Open data alone is not sufficient for inclusivity.”

→ The Rubin Science Platform provides a level playing field for interacting with Rubin data. Open-source software and open data are essential to open science and reproducibility.

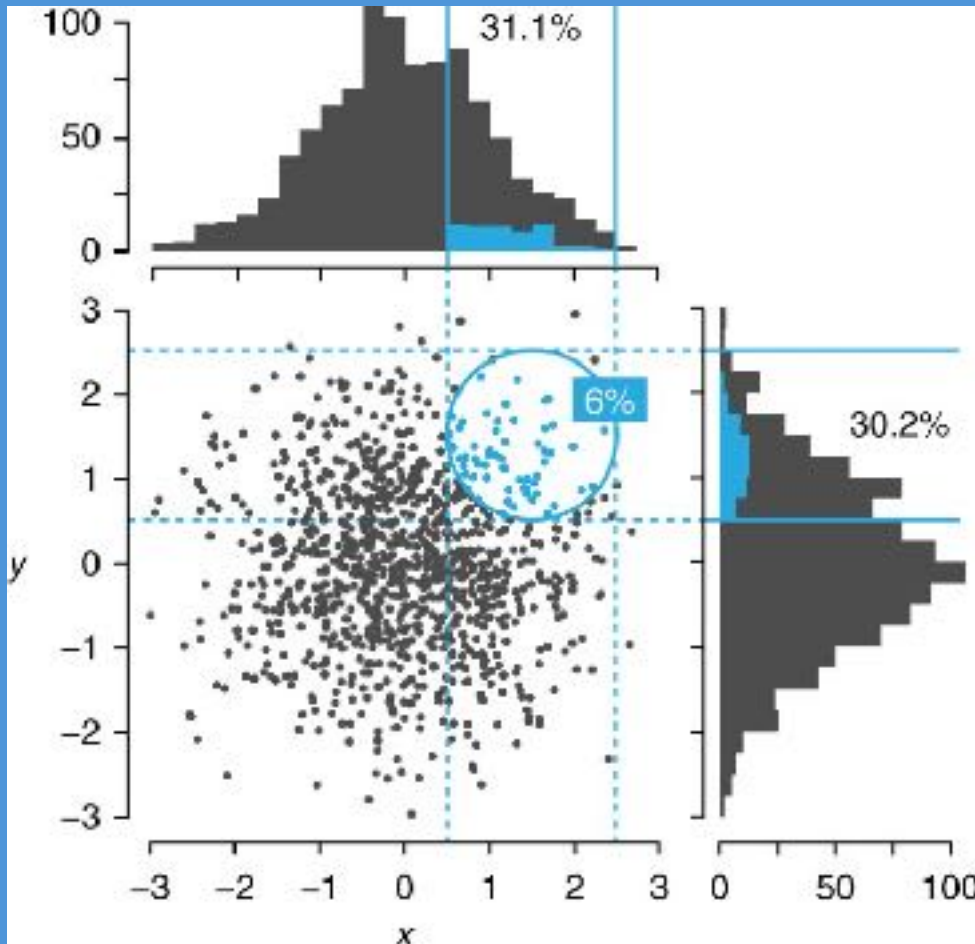
Democratizing Science

“Open data alone is not sufficient for inclusivity.”

- The Rubin Science Platform provides a level playing field for interacting with Rubin data. Open-source software and open data are essential to open science and reproducibility.
- Lowering the barrier to entry requires minimizing the investment (time, money, experience) necessary to meaningfully engage with the scientific questions that can be resolved with the data.

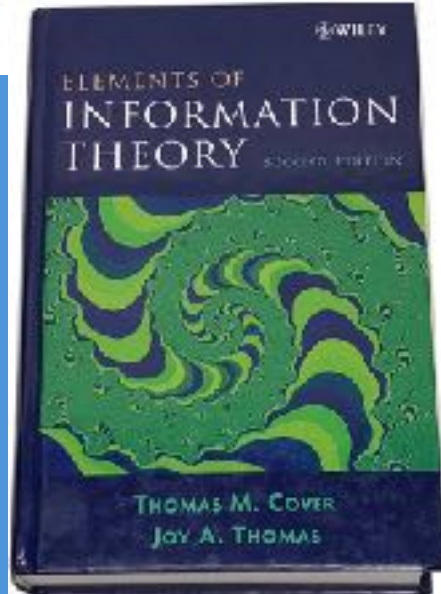
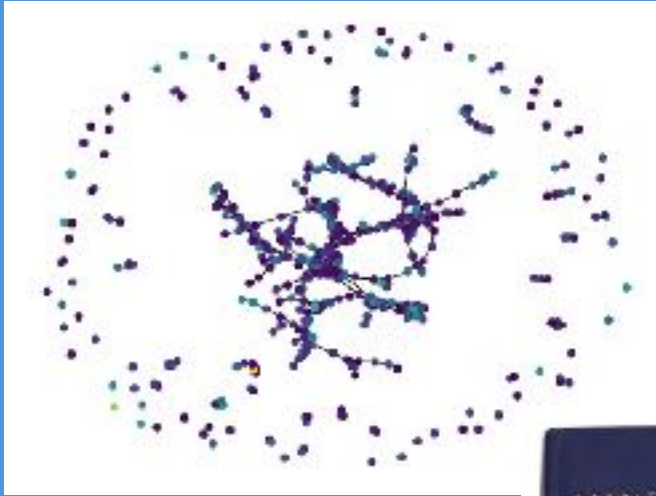
Challenge 2: Characterizing Structure in High Dimensions

Why is this hard? Statistics in high dimensions



More data is generally helpful in answering scientific questions, but as the dimensionality of a dataset increases, summary statistics become “less representative and harder to estimate”.

Possible solutions



Geometric, Physically Informed ML and Graphs – characterize the symmetries of high dimensional datasets

Information Theoretic Approaches – characterize conditional probabilities

Example: Correlation Explanation

Discovering Structure in High-Dimensional Data Through Correlation Explanation

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Abstract

We introduce a method to learn a hierarchy of successively more abstract representations of complex data based on optimizing an information-theoretic objective. Intuitively, the optimization searches for a set of latent factors that best explain the correlations in the data as measured by multivariate mutual information. The method is unsupervised, requires no model assumptions, and scales linearly with the number of variables which makes it an attractive approach for very high dimensional systems. We demonstrate that Correlation Explanation (CorEx) automatically discovers meaningful structure for data from diverse sources including personality tests, DNA, and human language.

See also: [gregversteeg/LinearCorex: Fast, linear version of CorEx for covariance estimation, dimensionality reduction, and subspace clustering with very under-sampled, high-dimensional data](#)

Challenge 3: Discovery in Large Datasets

“Needle in a haystack?”

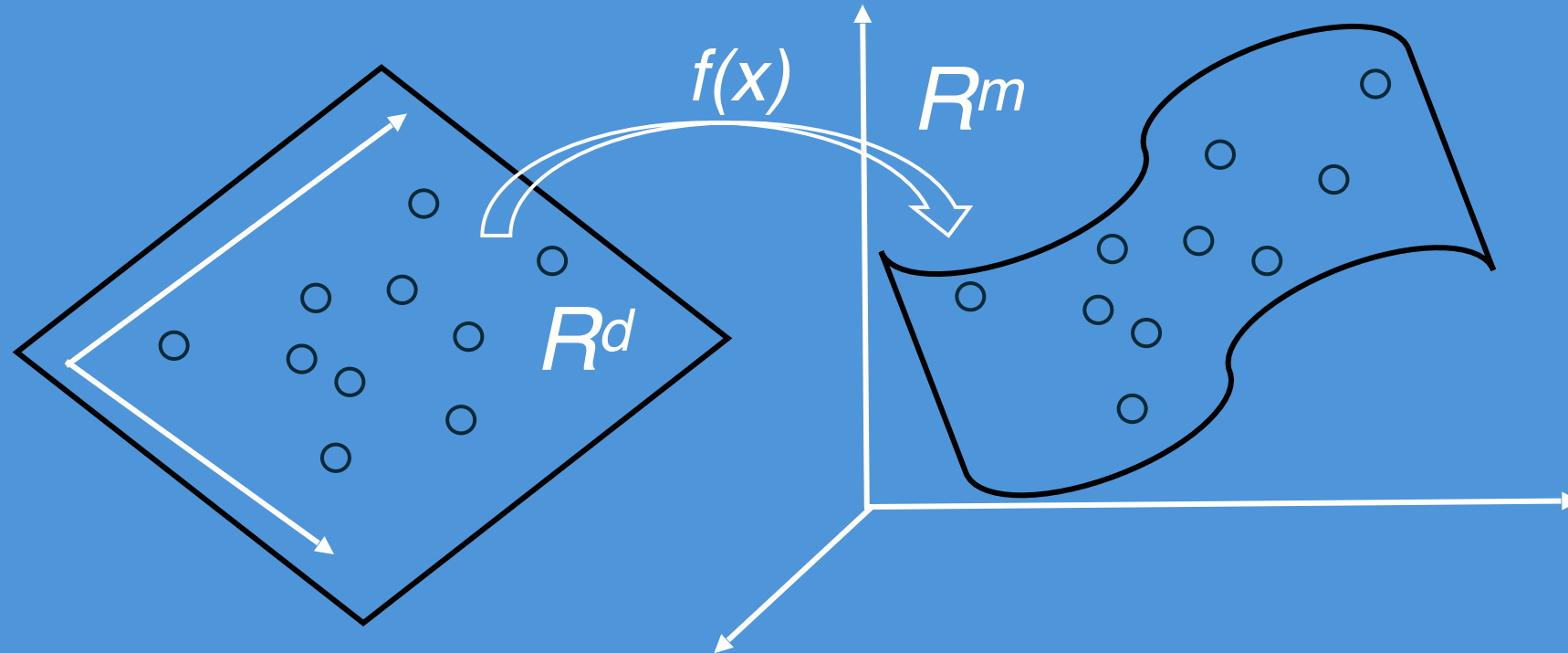


“Needle in a pile of needles”



anomaly detection (also referred to as ***outlier detection*** and sometimes as ***novelty detection***) is generally understood to be the identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well defined notion of normal behavior.

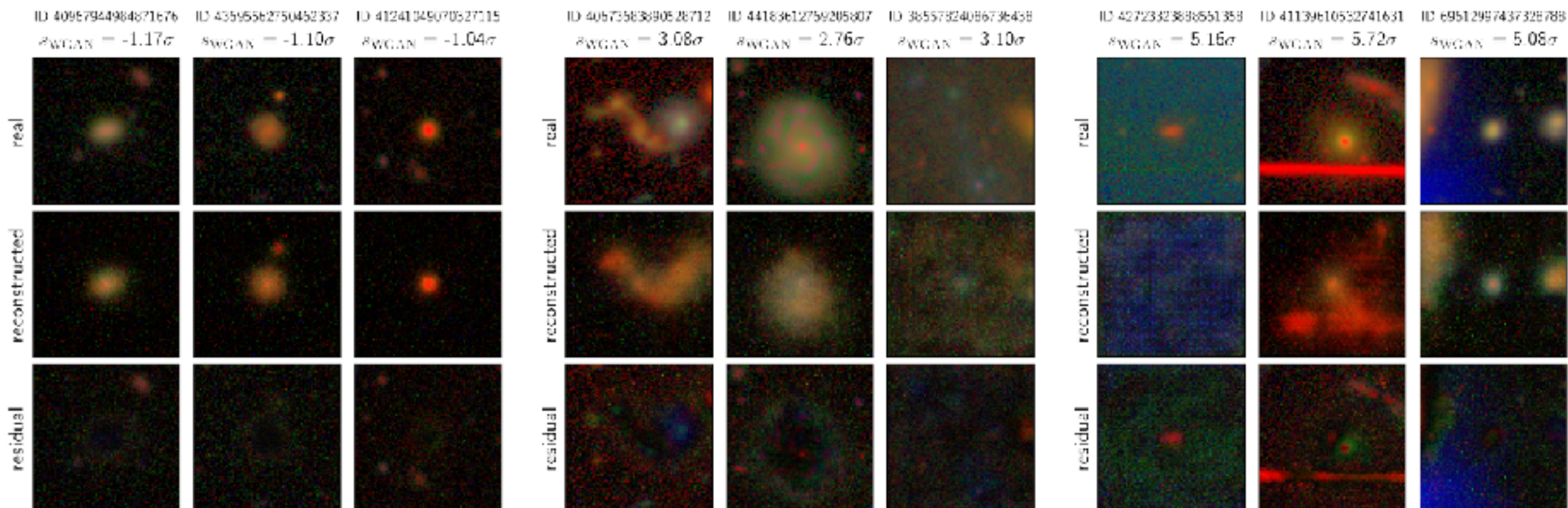
Representation Learning for Discovery



Discovery in representation learning is based on the intuitive idea that a member of a new or unusual class will look very different when projected onto the trained representation from the rest of the data.

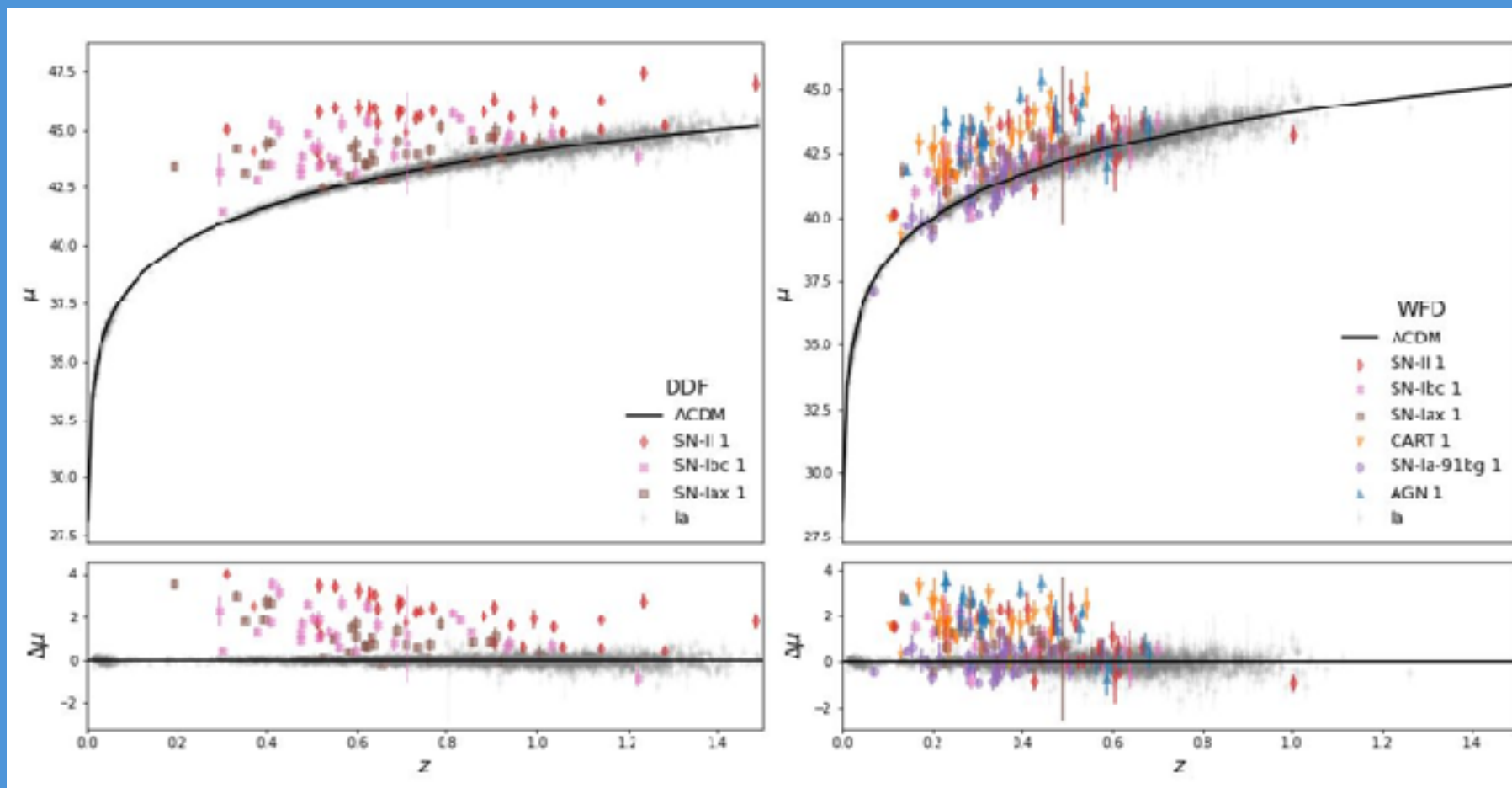
Most techniques are based on measuring distances in the representation.

Anomaly Detection with WGANs in HSC

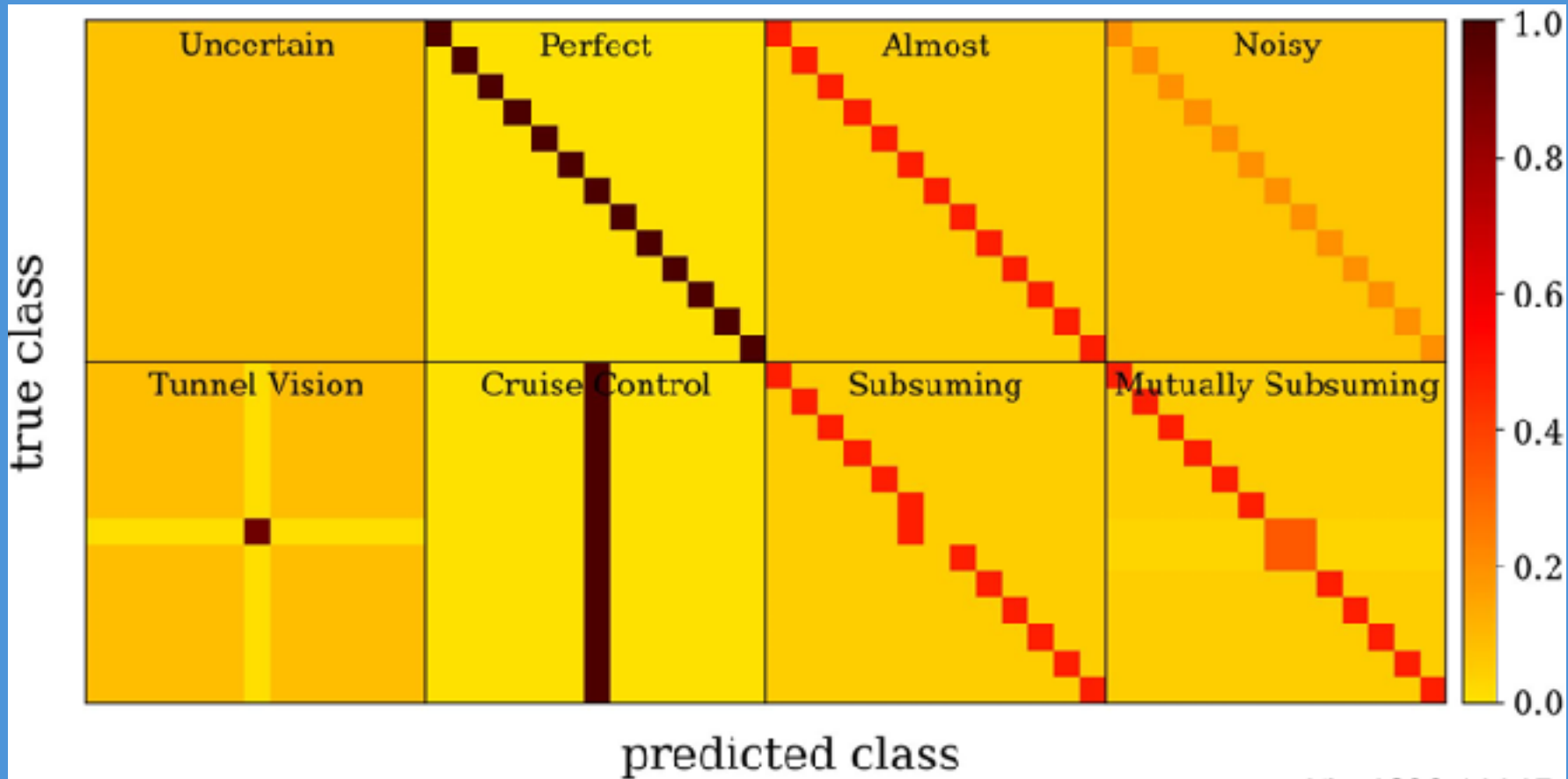


$$W(p_r(\mathbf{x}), p_z(\mathbf{z})) = \sup [\mathbb{E}_{\mathbf{x} \sim p_r(\mathbf{x})} [D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p_z(\mathbf{z})} [D(G(\mathbf{z}))]]$$

When and why do anomalies matter?



Classification Metrics

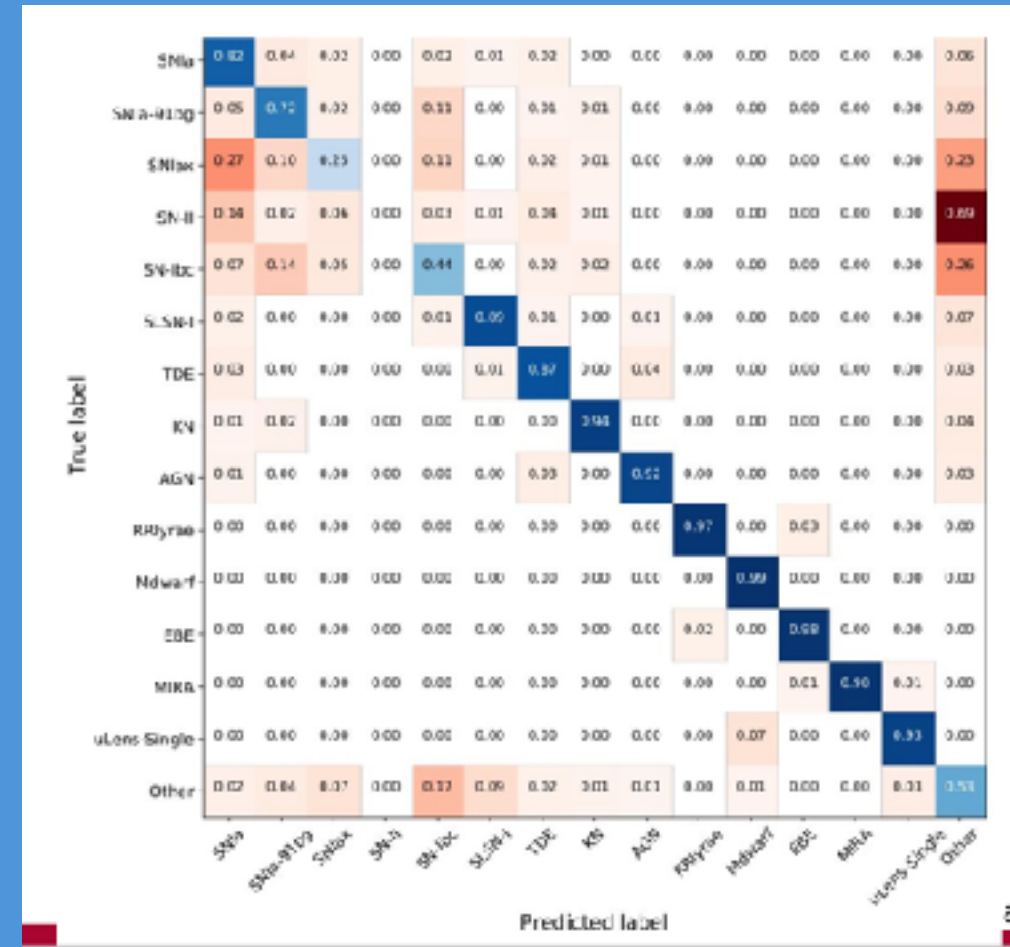


Classification Matrix

Suppose you have a method for assigning classifications to observations. By what metric do you judge the success of what the method returns?

First order answer: Confusion matrices.

Problem: Confusion matrices are (in general) degenerate! We can see that something has gone wrong but these metrics don't tell us how – makes discovery difficult!

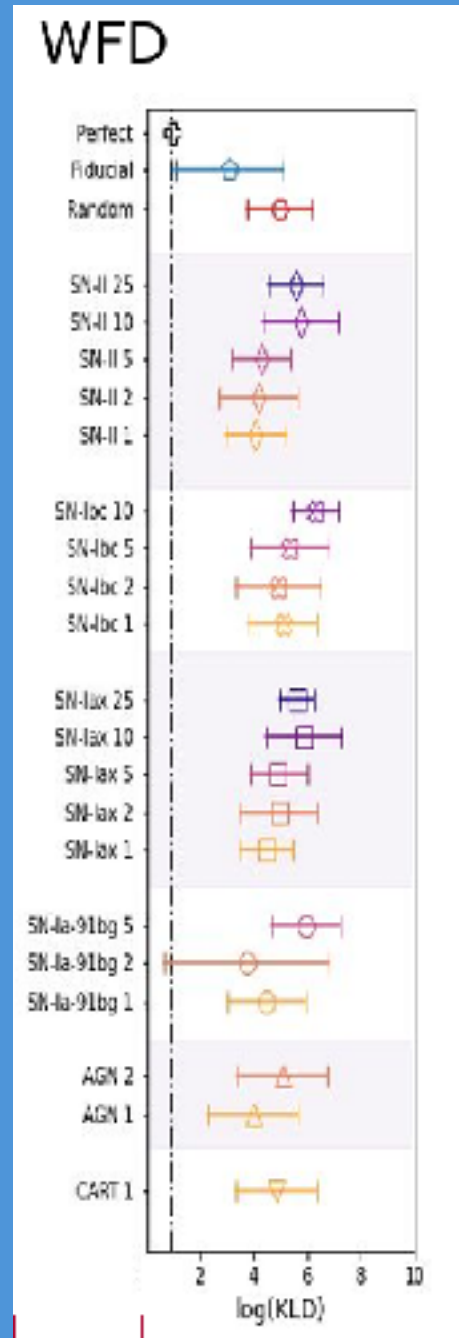


Distribution based metrics

The Kullback-Leibler Divergence is a (non-symmetric) distance measure between probability distributions.

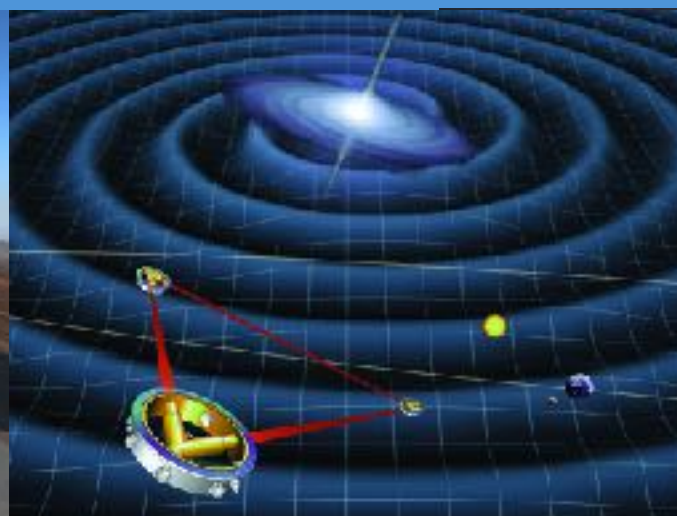
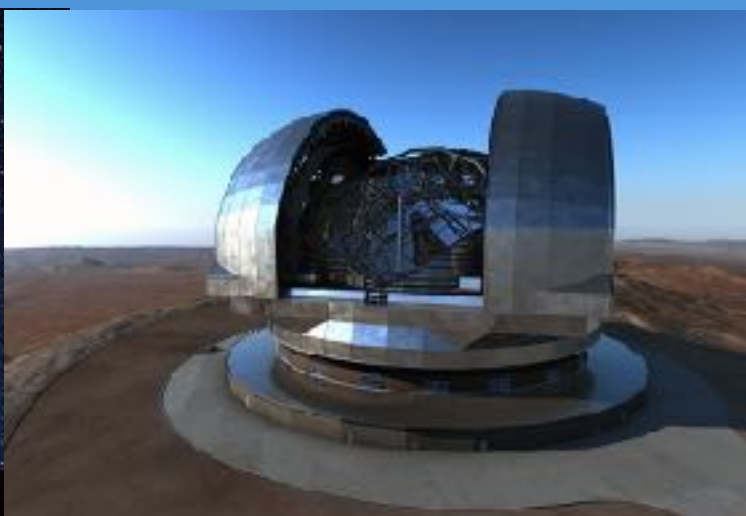
$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

Posterior based metrics can be more informative about the nature of classification failures.



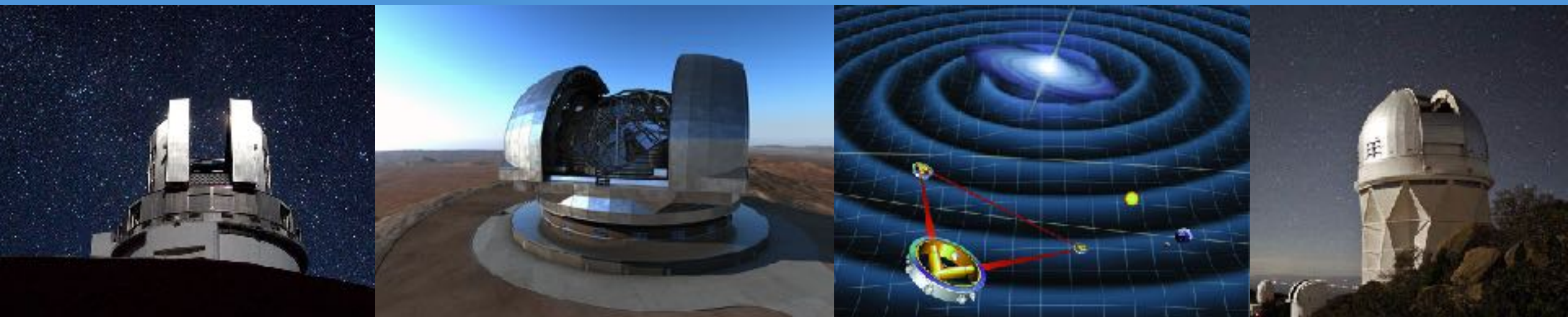
Malz et al., 2023

Challenge 4: Optimizing “Follow-Up” and Synergies with other facilities



Multi-band (or multi-probe in cosmology): Using two different parts of the same spectrum (photons, particles, gravitational waves)

Multi-messenger: Using two different modes of observation that are not the same carrier (photons, gravitational waves, or simulations)

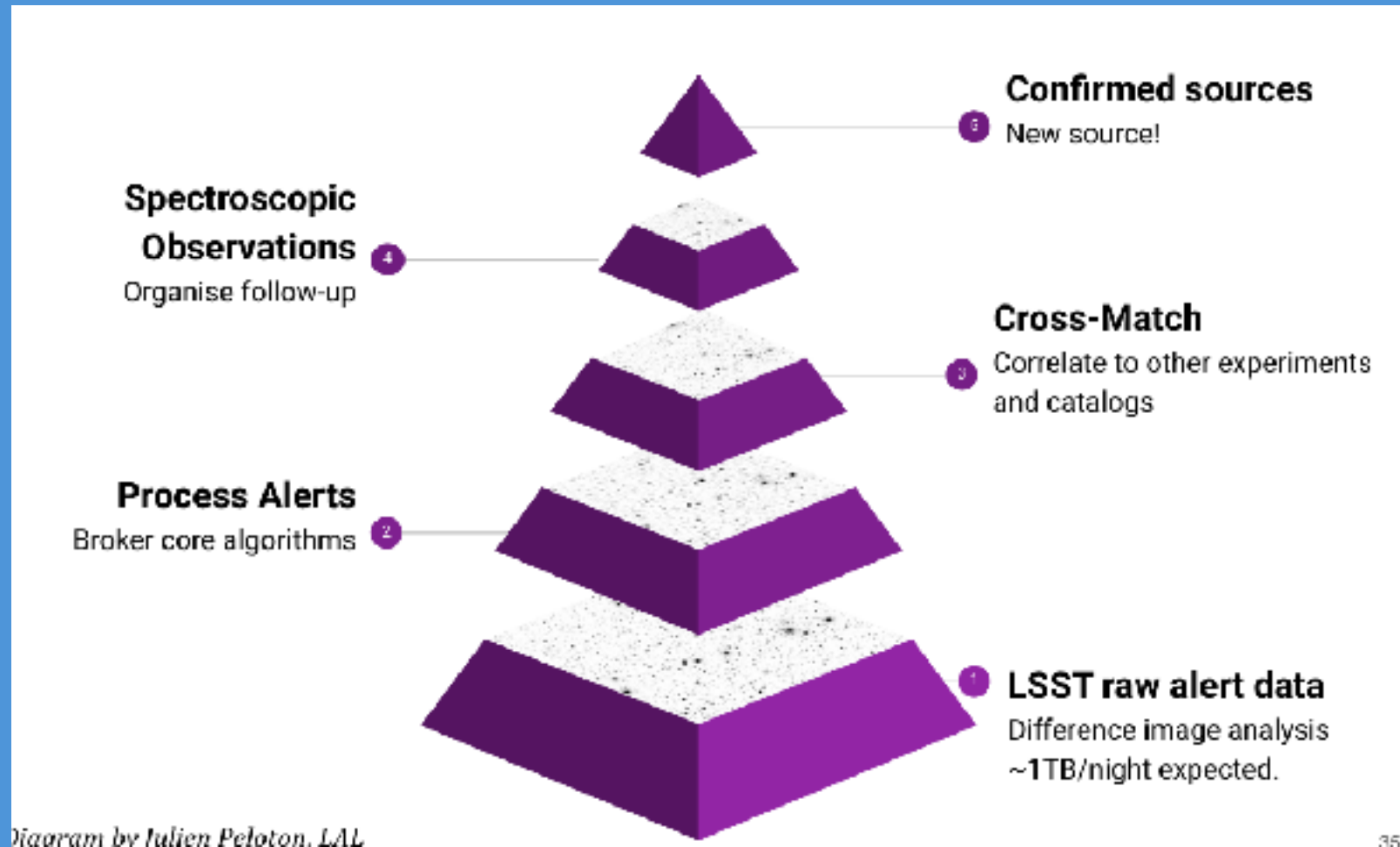


Multi-band (or multi-probe in cosmology): Using two different parts of the same spectrum (photons, particles, gravitational waves)

Multi-messenger: Using two different modes of observation that are not the same carrier (photons, gravitational waves, or simulations)

Data Challenge: How do we determine optimal tasking of limited resources for multi-band or multi-messenger follow-up?

Transient Follow up – Alert Brokers



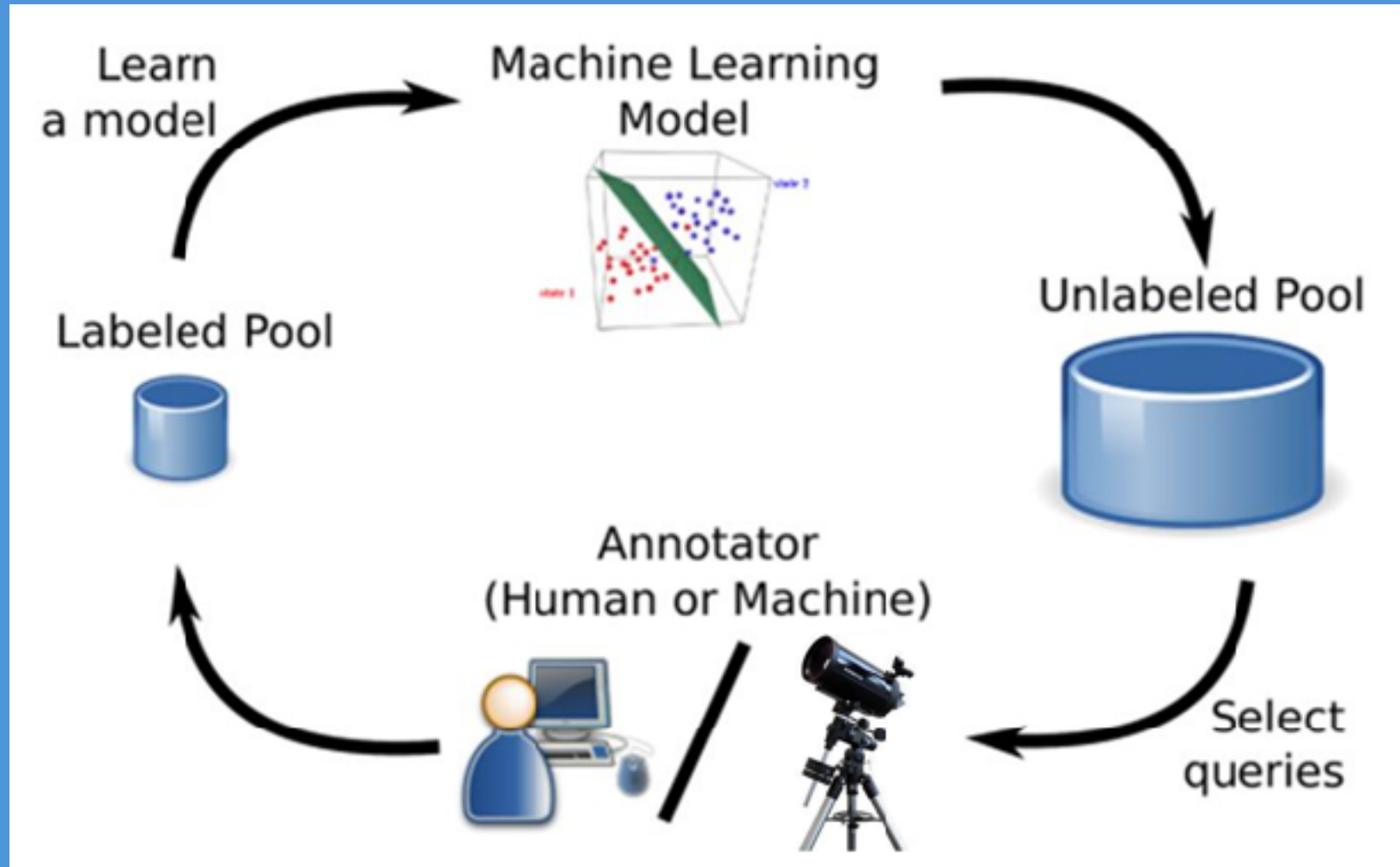
Transient Follow up – Alert Brokers



ALeRCE
Automatic Learning for the
Rapid Classification of Events



Active Learning for Follow-up



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Program Postdoctoral Scholar

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